MCQS

BT101 midterm preparation file

ZB

بسم الله الرحمن الرحيم

In the name of God , Most Gracious, Most merciful

BT101

MCQS

1. What is the basic unit of life?

a) Atom

b) Molecule

c) Cell

d) Tissue

Answer: c) Cell

2. Which macromolecule is the main source of energy for living organisms?

a) Lipids

b) Proteins

c) Carbohydrates

d) Nucleic acids

Answer: c) Carbohydrates

3. Who is known as the father of modern biology?

a) Aristotle

b) Louis Pasteur

c) Darwin

d) Mendel

Answer: a) Aristotle

4. Which organelle is responsible for cellular respiration?

a) Chloroplast

b) Nucleus

c) Mitochondria

d) Golgi apparatus

Answer: c) Mitochondria

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5. DNA is made up of repeating units called:

a) Amino acids

b) Nucleotides

c) Monosaccharides

d) Fatty acids

Answer: b) Nucleotides

6. What type of bond holds water molecules together?

a) Covalent bond

b) Ionic bond

c) Hydrogen bond

d) Metallic bond

Answer: c) Hydrogen bond

7. Which scientist discovered the structure of DNA?

a) Watson and Crick

b) Mendel

c) Hooke

d) Rosalind Franklin

Answer: a) Watson and Crick

8. The plasma membrane is made up of:

a) Proteins and carbohydrates

b) Lipids and proteins

c) Nucleotides and lipids

d) Carbohydrates and nucleotides

Answer: b) Lipids and proteins

9. What is the function of ribosomes?

a) DNA replication

b) Protein synthesis

c) Lipid synthesis

d) Energy production

Answer: b) Protein synthesis

10. Which of the following is a prokaryote?

a) Fungi

b) Algae

c) Bacteria

d) Protozoa

Answer: c) Bacteria

11. Which molecule stores genetic information?

a) RNA

b) DNA

c) Protein

d) Lipid

Answer: b) DNA

12. What is the main component of the cytoplasm?

a) Lipids

b) Water

c) Proteins

d) Carbohydrates

Answer: b) Water

13. What does ATP stand for?

a) Adenosine Triphosphate

b) Adenine Tetraphosphate

c) Amino Triphosphate

d) Adenosine Diphosphate

Answer: a) Adenosine Triphosphate

14. Which organelle is known as the "powerhouse" of the cell?

a) Nucleus

b) Chloroplast

c) Mitochondria

d) Lysosome

Answer: c) Mitochondria

15. Enzymes are made of:

a) Nucleic acids

b) Proteins

c) Lipids

d) Carbohydrates

Answer: b) Proteins

16. The pH scale ranges from:

a) 1 to 10

b) 1 to 14

c) 0 to 7

d) 0 to 14

Answer: d) 0 to 14

17. Which of the following is not a macromolecule?

a) Lipids

b) Vitamins

c) Proteins

d) Nucleic acids

Answer: b) Vitamins

18. Which organelle is responsible for photosynthesis?

a) Chloroplast

b) Mitochondria

c) Ribosome

d) Golgi apparatus

Answer: a) Chloroplast

19. What are the subunits of proteins?

a) Nucleotides

b) Amino acids

c) Fatty acids

d) Monosaccharides

Answer: b) Amino acids

20. Which is an example of passive transport?

a) Endocytosis

b) Osmosis

c) Exocytosis

d) ATP synthesis

Answer: b) Osmosis

21. What is the function of lysosomes?

a) Photosynthesis

b) Digestion of cellular waste

c) Protein synthesis

d) DNA replication

Answer: b) Digestion of cellular waste

22. Which of the following is a double-membraned organelle?

a) Ribosome

b) Mitochondria

c) Centriole

d) Cytoskeleton

Answer: b) Mitochondria

23. Which sugar is found in RNA?

a) Ribose

b) Glucose

c) Deoxyribose

d) Fructose

Answer: a) Ribose

24. The main structural component of plant cell walls is:

a) Cellulose

b) Protein

c) Starch

d) Glycogen

Answer: a) Cellulose

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25. Which process produces the most ATP?

a) Glycolysis

b) Fermentation

c) Krebs cycle

d) Electron transport chain

Answer: d) Electron transport chain

26. What is the pH of pure water?

a) 5

b) 7

c) 9

d) 14

Answer: b) 7

27. The first scientist to observe cells was:

a) Hooke

b) Leeuwenhoek

c) Pasteur

d) Virchow

Answer: a) Hooke

28. Which type of RNA carries amino acids to ribosomes?

a) mRNA

b) tRNA

c) rRNA

d) snRNA

Answer: b) tRNA

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29. What is the main function of the Golgi apparatus?

a) Protein synthesis

b) Modifying and packaging proteins

c) Cellular respiration

d) DNA replication

Answer: b) Modifying and packaging proteins

30. Which organelle is involved in lipid synthesis?

a) Rough ER

b) Smooth ER

c) Ribosome

d) Golgi apparatus

Answer: b) Smooth ER

31. What is the function of chlorophyll?

a) Cellular respiration

b) Absorb sunlight for photosynthesis

c) Protein synthesis

d) DNA replication

Answer: b) Absorb sunlight for photosynthesis

32. The fluid mosaic model describes the structure of:

a) Cell wall

b) Plasma membrane

c) Cytoplasm

d) Nucleus

Answer: b) Plasma membrane

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33. Which of the following is a lipid?

a) Glucose

b) Glycogen

c) Cholesterol

d) Cellulose

Answer: c) Cholesterol

34. What is the role of centrioles in the cell?

a) Photosynthesis

b) Protein synthesis

c) Cell division

d) Energy production

Answer: c) Cell division

35. Which phase of mitosis involves chromosomes aligning at the center?

a) Prophase

b) Metaphase

c) Anaphase

d) Telophase

Answer: b) Metaphase

36. What is the basic structure of a nucleotide?

a) Sugar, phosphate, base

b) Sugar, lipid, base

c) Amino acid, base, sugar

d) Phosphate, protein, sugar

Answer: a) Sugar, phosphate, base

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37. Which of the following is an example of active transport?

a) Osmosis

b) Diffusion

c) Sodium-potassium pump

d) Facilitated diffusion

Answer: c) Sodium-potassium pump

38. What is the primary function of the rough ER?

a) Lipid synthesis

b) Protein synthesis

c) ATP production

d) Cell division

Answer: b) Protein synthesis

39. The basic unit of heredity is called:

a) DNA

b) Gene

c) Chromosome

d) Protein

Answer: b) Gene

40. Which organelle contains digestive enzymes?

a) Lysosome

b) Golgi apparatus

c) Ribosome

d) Mitochondria

Answer: a) Lysosome

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41. Which type of bond is strongest in biological molecules?

a) Hydrogen bond

b) Ionic bond

c) Covalent bond

d) Metallic bond

Answer: c) Covalent bond

42. The process by which plants make their own food is called:

a) Respiration

b) Photosynthesis

c) Fermentation

d) Transpiration

Answer: b) Photosynthesis

43. The genetic material of prokaryotic cells is found in:

a) Nucleus

b) Cytoplasm

c) Nucleoid region

d) Ribosomes

Answer: c) Nucleoid region

44. Which organelle is responsible for detoxifying harmful substances?

a) Rough ER

b) Smooth ER

c) Golgi apparatus

d) Lysosome

Answer: b) Smooth ER

45. Which phase of the cell cycle involves DNA replication?

a) G1 phase

b) S phase

c) G2 phase

d) M phase

Answer: b) S phase

46. What is the primary function of carbohydrates in cells?

a) Storage of genetic information

b) Long-term energy storage

c) Quick energy source

d) Structural support in membranes

Answer: c) Quick energy source

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WhatsApp:

https://whatsapp.com/channel/0029VaODheTDp2Q8YVcoON2p

Second WhatsApp channel

channel on WhatsApp:

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https://youtu.be/-AMj4iNz0UE?si=hg29pJZ1kujzg57J

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Passing criteria of this course

Letter Grade	Grade Points	Equivalent Percentage
A+	4.00	90-100
A	4.00	85-89
A-	3.66 - 3.99	80-84
B+	3.33 - 3.65	75-79
B	3.00 - 3.32	71-74
B-	2.66 - 2.99	68-70
C	2.00 - 2.65	61-67
D	1.00 - 1.99	50-60
F	0.00	00-49

Passing Criteria for a Course:

The passing criteria is defined in such a way that it ensures the student shall be consistent in his studies throughout the semester.
Therefore, for passing a course/subject, student shall fulfill the following:
a) Secure minimum 20% score in Formative Assessments/Mid terms
b) Secure minimum 20% score in Final Term Examinations.
c) Secure at least 40% marks in aggregate while fulfilling the above requirements

Course Selection and Credit Hours

Q: How do I select courses when the course selection is open?
A: Follow these steps:
1. Check the credit hours allowed by your university (e.g., 21, 18, or 15 credit hours).
2. Each subject typically has 3 credit hours.
3. Divide the total allowed credit hours by 3 to determine how many subjects you can select.
Examples:
21 credit hours = 7 su18 credit hours = 6 subjects
15 credit hours = 5 subjects
Remember, the university may allow different credit hours for each student, so check your specific allowance.
By following these steps, you can make informed decisions during course selection and manage your credit hours effectively.

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Advice

Hmaesha relax ho ke parha kryn panic mat hua kryn apky parents ky bohat khawab hoty hn wo pura krny ki puri koshish kea kryn kbi b ksi pe depend na kea kryn apko bs ak insan success kr saqta ha or wo insan ap khud hn apky elawa koi nahe ap bs koshish kryn Allah pak pe strong yakeen rakha kryn or mehnat krke sb Allah ke hawaly kr dea kryn everything is possible be brave be strong stay blessed
Hum insan hamesha moat se darty or bhagty hn or moat se bachny ki koshish krty hn jabke moat ny ana hi ana ha hum

moat se nahe bach saqty humy jahanam se bachny ki koshish krni chaheay hum jahanam se bach saqty hn
Insan ko 3 chezo se dar lagta ha Moat Risk/Dolat Ezat Shohrat Fame

Moat

Humy pta hona chaheay Moat tab ani ha jab Allah chahy us se pehly puri dunya bi ak tararf ho ke apko marna chahy to apka kuch nahe beggar saqti

Risk/Dolat

Risk Dolat sb Allah pak ke hath m ha puri dunya ak taraf b ho jay na apsy apka risk cheen saqti ha na de saqti ha Ye sirf ALLAh pak ky hath m ha

Izat Zilat shohrat Fame

Izat Zilat Sirf Allah de saqty hn puri dunya mel ke b apko zra brabr b demage nahe kr saqti Agr ALLAH

Apko izat dena chahay to puri dunya ak tararf ho ke b apko 1% b nuksan nahe pohancha saqti
Phr Dar ks bat ka ??????????????
Be brave be strong Just put your trust To ALLAH
Hamesha Confident or himat se raho kbi ksi k samne mat jhuko puri dunya mel k b apka kuch b nahe beggar saqti

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MY REQUEST FROM ALL OF YOU

ZB Request from All of you My family
Mjy ap sbki help or support ki zarort ha or wo ye k hum sb Mel kr Allah ka Quran ki Urdu translation logo tak pohanchy or Quran ko samjna asaaan kryn dosro k leay sb tak Allah ka Quran pohanchy it's my compaign hum log 70+ age k ho jaty hn phr b hmy namaz tak ki translation nahe pta hoti k hum Allah pak se

Kya Dua kr rhy hn so hum youngster's ko Mel kr puri takat or energy ke sath Allah ka Quran spread krna ha
Hum puri Koshish kr rhy hn k hum sb Mel k Quran ki translation or most important topics ko maximum share kry with translation hmara maksad Quran ko spread krna ha be a part of us
m apni pocket se ye sb kr rha ho Allah ka Quran spread krny k leay apne or mene sbny Marna ha ak din to q na Allah k leay Kuch kryn apni energies apni power ko bajay negative use krny k Allah k leay invest kryn khud ko be a part of us
Agr m 23 ki age m apny sare sources use kr rha ho pocket se heavy amounts give away kr rha ho Quran spread krny k leay to ap just share to kr saqty hn itna e krdyn

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Agr ap mujsy contact na kr sky too b ap jo b hn jaha bi hn waha Allah ka quran pheelay zada se zada with urdu translation jitni himat ha utna share kryn

MY Family

ust Relax and focus! Exams are not difficult. Put in your 100% effort and trust in Allah.

AL-Quran

"And indeed, with hardship comes ease." (Quran 94:5)
Remember, I'm here to support you! Stay brave and strong!"

May Allah bless you